

METR · MARCH 2025

Measuring AI Ability to Complete Long Software Tasks

50% task-completion time horizon doubling every ~7 months

Kwa, West, Becker, et al. · arXiv:2503.14499

METHODOLOGY

A New Metric: 50% Task-Completion Time Horizon

01 Assemble Task Suite

170 diverse software and research tasks spanning 1 second to 8 hours of skilled human effort.

02 Measure & Compare

Time skilled humans on each task, then measure AI agent success rates under identical conditions.

CENTRAL FINDING

AI Task Horizon Grew from 2 Seconds to ~1 Hour 50 Minutes in 6 Years

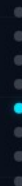
Exponential growth in the 50% time horizon from GPT-2 (2019) to o3 (2025).



MODEL COMPARISON

Success Probability Drops Sharply as Task Length Increases

Frontier models push the curve rightward — later models maintain higher success on longer tasks.



CAPABILITY ANALYSIS

Reasoning, Tool Use, and Error Recovery Drive Horizon Growth

Four interlocking capabilities explain the exponential expansion in autonomous task completion.

Improved Logical Reasoning

Models plan multi-step solutions more effectively, decomposing complex objectives into executable sub-tasks.

Reliability & Self-Awareness

Agents increasingly know when they are stuck or uncertain, enabling safer delegation and halting.

Better Tool Use

More reliable invocation of code interpreters, search, file systems, and APIs to extend capability.

FORWARD PROJECTION

Extrapolation Predicts Month- Long Task Automation by 2028–2031

If the exponential trend continues, frontier AI would autonomously complete tasks requiring a month of skilled human labor.

SAFETY GOVERNANCE

Exponential Autonomous Capability Growth Demands Proactive Safety Governance

Five takeaways for AI safety researchers, frontier lab leadership, and policy analysts.

